

Summary Data on Acid Cropland and Lime Requirements in Rwanda

GAIA

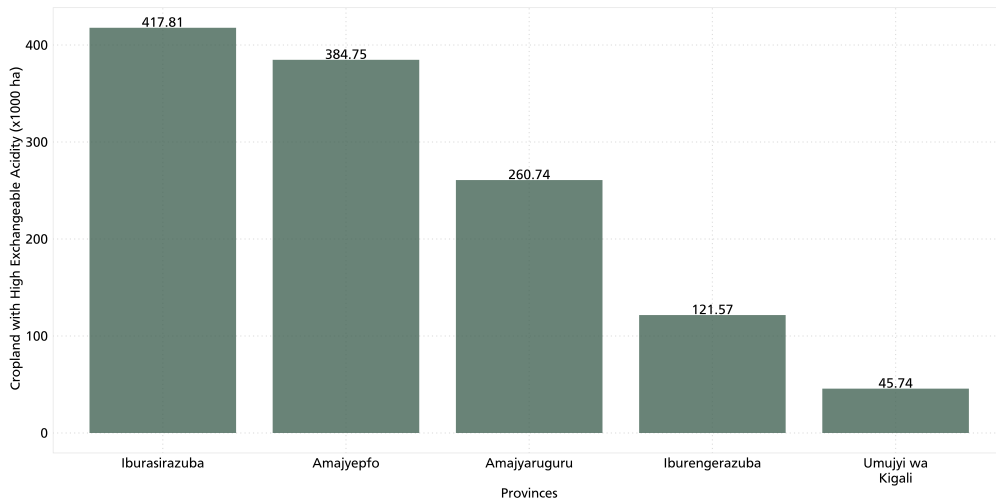
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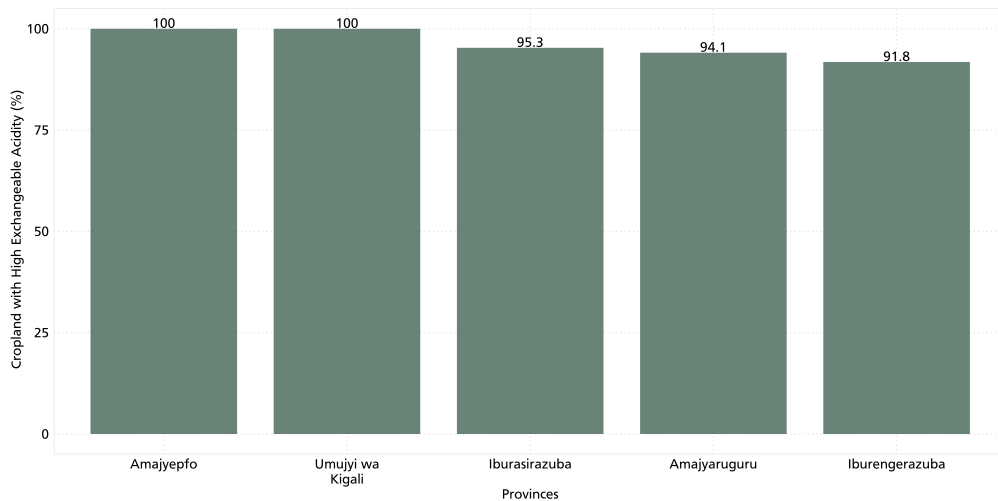
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- 3 Yield Loss
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Top 10 Provinces in Rwanda with the highest acidic cropland



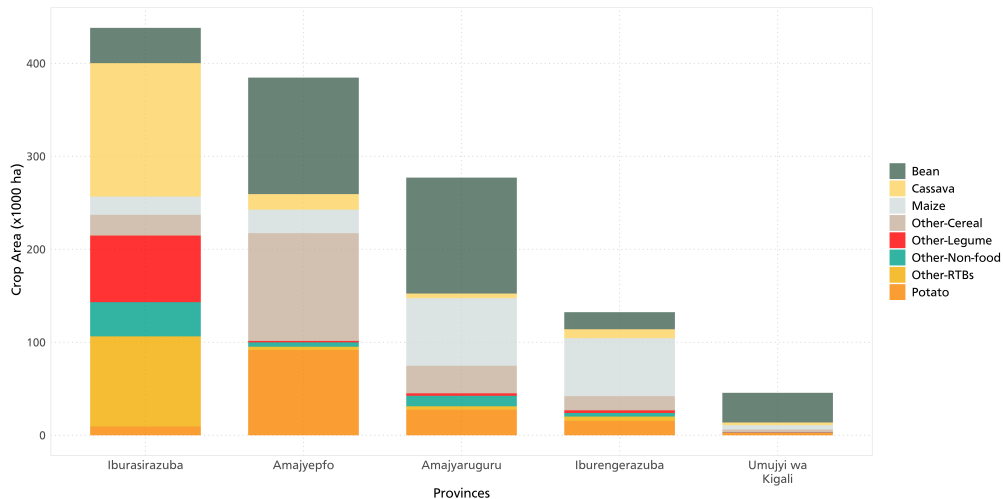
Top 10 Provinces in Rwanda with the highest acidic cropland (Percentage of total cropland)



└ Total crop area distribution (ha)

└ Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland

Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland



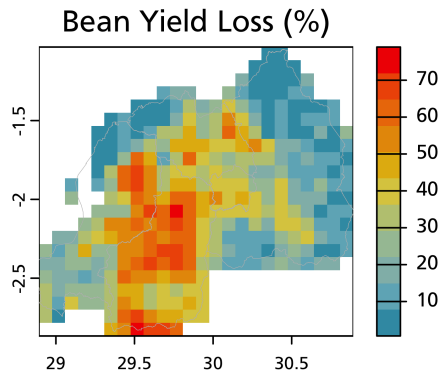
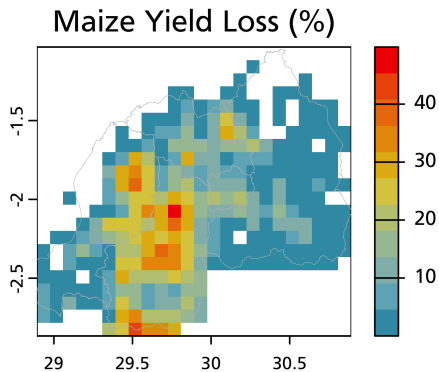
Rwanda–Summary

└ Total crop area distribution (ha)

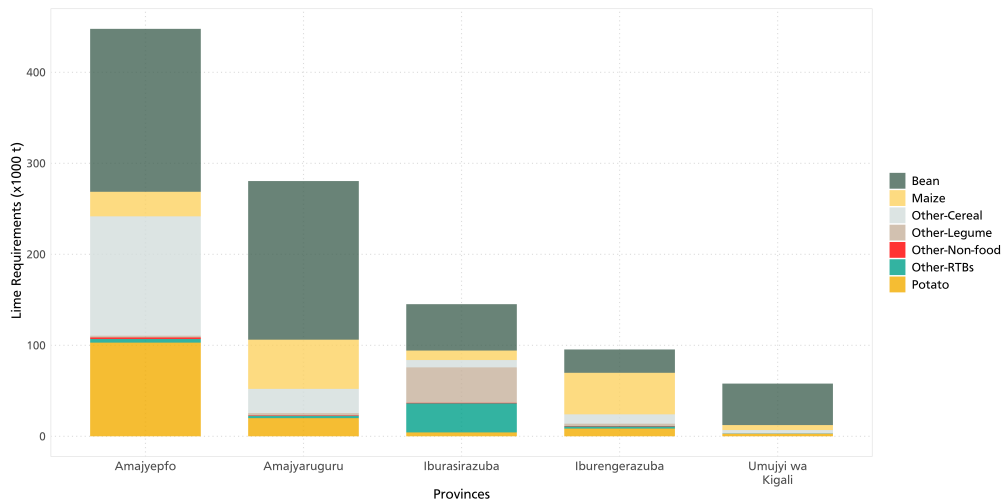
└ Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland

Provinces	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Amajyaruguru	124.66	5.03	72.61	29.67	2.67	11.32	3.68	27.53
Amajyepfo	125.33	16.69	25.34	115.82	1.59	4.70	3.50	91.79
Iburasirazuba	37.96	143.45	19.67	22.34	71.59	36.79	97.00	9.47
Iburengerazuba	18.36	9.56	62.46	15.18	2.88	3.87	4.47	15.66
Umujyi wa Kigali	31.90	2.93	4.72	3.03	0.36	0.26	0.16	2.39

Yield loss due to soil acidity for Maize and Bean in Rwanda



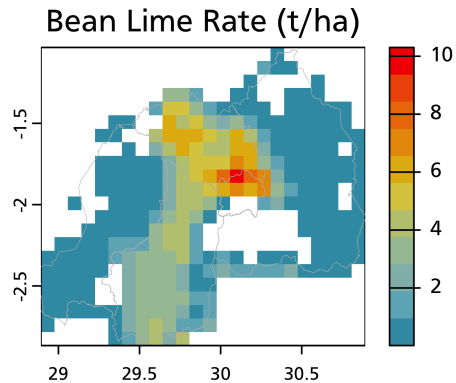
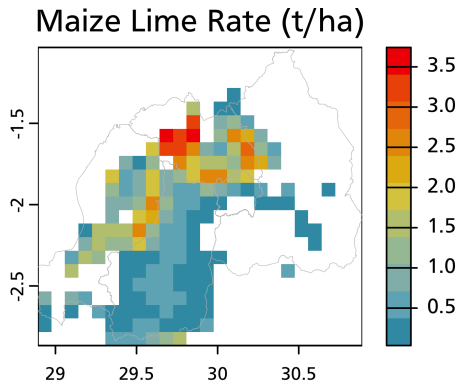
Lime requirements (t) for acid soil remediation in Rwanda



Lime requirements (t) for acid soil remediation in Rwanda (table)

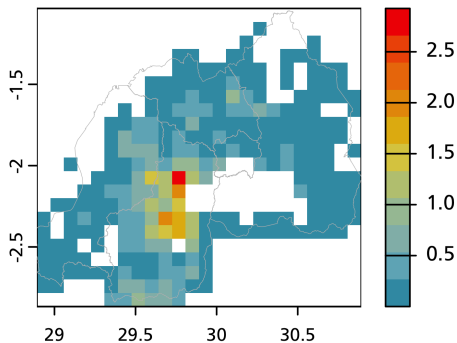
Provinces	Bean	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Amajyaruguru	174.32	54.03	26.63	2.49	0.56	2.48	20.00
Amajyepfo	179.04	27.02	131.15	1.93	1.70	3.98	102.97
Iburasirazuba	50.99	10.44	8.09	38.94	0.51	31.99	4.31
Iburengerazuba	25.65	45.60	10.35	2.49	0.46	2.44	8.46
Umujyi wa Kigali	45.57	5.44	3.49	0.49	0.01	0.19	2.75

Lime rate for Maize and Bean in Rwanda

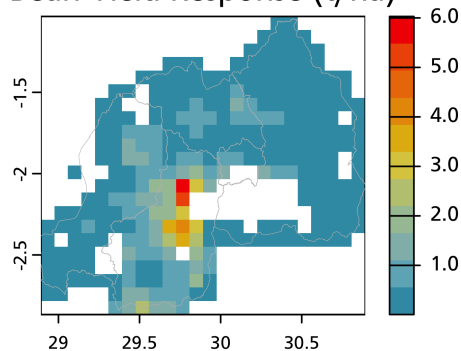


Yield response to lime application in Rwanda

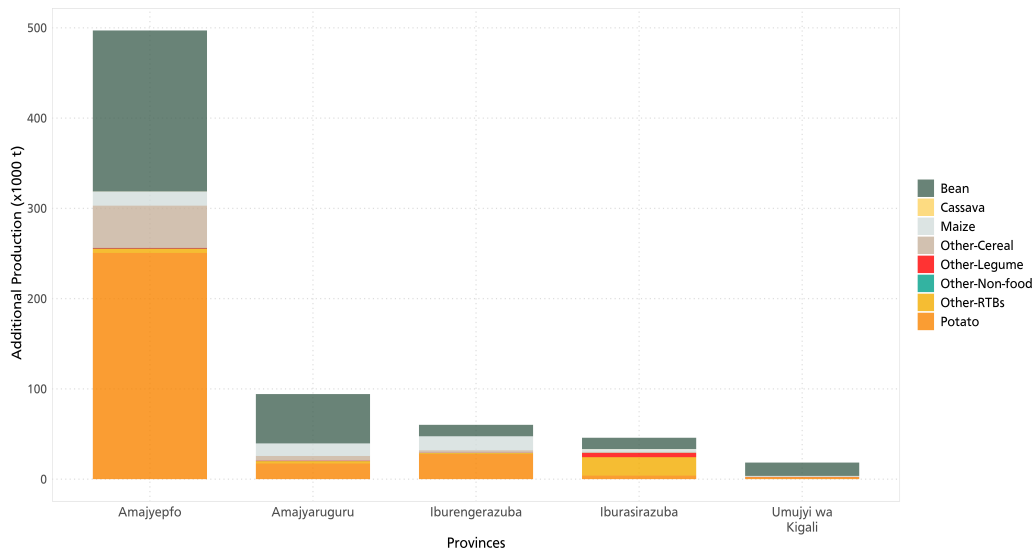
Maize Yield Response (t/ha)



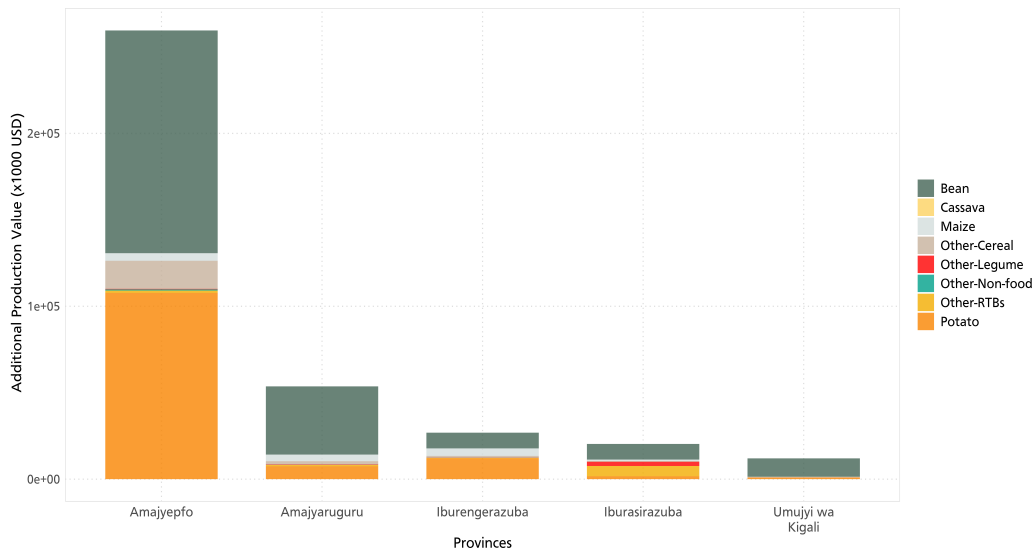
Bean Yield Response (t/ha)



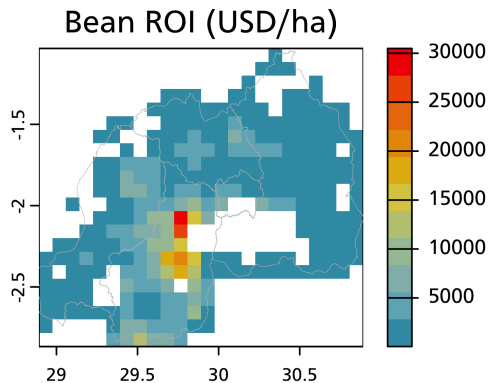
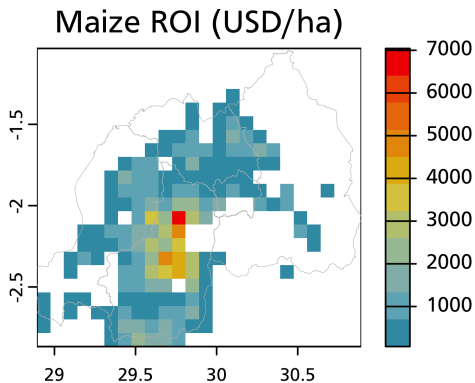
Potential for additional production (tone) in top 10 acidic Provinces in Rwanda



Potential for additional production value (US\$) in top 10 acidic Provinces in Rwanda

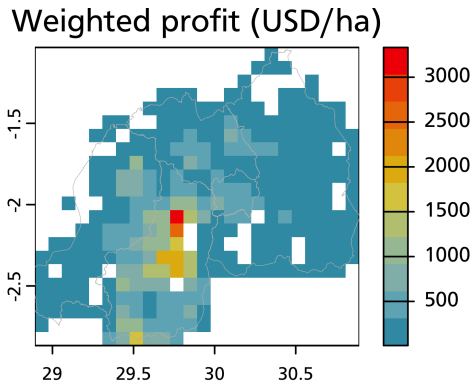
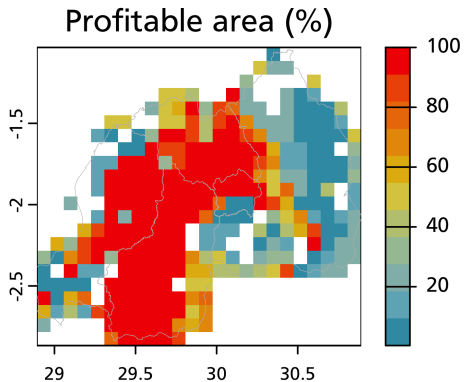


Return on investment for lime application in Rwanda



- └ Profitability of Lime Application
- └ Profitability of lime application in Rwanda

Profitability of lime application in Rwanda



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

For details, visit www.acidsoils.africa.